RESEARCH ARTICLE | JULY 13 2018

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Special Collection: Chemical Physics of Charged Macromolecules

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J. Chem. Phys. 149, 163308 (2018)
https://doi.org/10.1063/1.5029268

THE JOURNAL OF CHEMICAL PHYSICS **149**, 163308 (2018)

Polyelectrolyte complex coacervation by electrostatic dipolar interactions

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(Received 12 March 2018; accepted 21 May 2018; published online 13 July 2018)

We address complex coacervation, the liquid-liquid phase separation of a solution of oppositely charged polyelectrolyte chains into a polyelectrolyte rich complex coacervate phase and a dilute aqueous phase, based on the general premise of spontaneous formation of polycation-polyanion complexes even in the homogeneous phase. The complexes are treated as flexible chains made of dipolar segments and uniformly charged segments. Using a mean field theory that accounts for the entropy of all dissociated ions in the system, electrostatic interactions among dipolar and charged segments of complexes and uncomplexed polyelectrolytes, and polymer-solvent hydrophobicity, we have computed coacervate phase diagrams in terms of polyelectrolyte composition, added salt concentration, and temperature. For moderately hydrophobic polyelectrolytes in water at room temperature, neither hydrophobicity nor electrostatics alone is strong enough to cause phase separation, but their combined effect results in phase separation, arising from the enhancement of effective hydrophobicity by dipolar attractions. The computed phase diagrams capture key experimental observations including the suppression of complex coacervation due to increases in salt concentration, temperature, and polycation-polyanion composition asymmetry, and its promotion by increasing the chain length, and the preferential partitioning of salt into the polyelectrolyte dilute phase. We also provide new predictions such as the emergence of loops of instability with two critical points. *Published by AIP Publishing.* https://doi.org/10.1063/1.5029268

I. INTRODUCTION

Mixtures of oppositely charged polyelectrolytes in aqueous solutions readily undergo phase separation resulting in a coexistence of a polymer-rich phase, commonly called the complex coacervate phase, and a polymer-poor phase.[1–8] Starting from the initial report by de Jong and Kruyt[1] over 80 years ago, there have been numerous investigations on this subject driven by applications such as therapeutic drug delivery,[9] coatings,[10] underwater adhesives,[11,12] etc., and attempts[2–8,13–49] to understand the phenomenon. The phase behaviors of solutions of oppositely charged polyelectrolytes are rich due to the multi-component nature (two kinds of polyelectrolytes, salt, and water) of the system and long-ranged electrostatic interactions among the various components. Generally, the thermodynamic phase stability of these systems depends on various factors including temperature, salt concentration, the polyelectrolyte molecular weight, polycation-polyanion composition asymmetry, and polyelectrolyte backbone charge density. The underlying physics of complex coacervation lies in the interplay of entropy and enthalpy due to both short-ranged Van der Waals interactions and long-ranged electrostatics. There have been more recent reviews[50–52] on this subject summarizing various experimental[4–8,13–34] and theoretical results.[35–49] Despite such intense investigations on complex coacervation, this phenomenon still remains as yet to be fully understood. In this paper, the focus is on the theoretical aspects of complex coacervate phase behavior and, in particular, on a new theory[52] based on electrostatic dipolar interactions between associated polycation-polyanion structures.

The earliest theory of coacervate phase behavior is the Voorn-Overbeek theory (VOT).[2,3] The basic ingredients of VOT are the entropy of mixing of chains (by following the Flory-Huggins theory) and the electrostatic correlations of charged monomers (using the Debye-Hückel theory of simple electrolytes) by assuming that all polycations and polyanions in the system are completely decomposed into their charged monomers. The role of counterions of the polyelectrolytes, in terms of how they enter into the Debye screening length and translational entropy, is not adequately described in the original VOT. Beyond the issue of such a description and the lack of electrostatic correlation of polyelectrolyte chains in the VOT, the Debye-Hückel approximation has its well-known limitations. In spite of these drastic and well-known approximations, the VOT is found to be in good agreement with some experiments. Attempts based on theory and simulations have been made in order to understand why such a crude model even works and to improve the approximations in the VOT. The various attempts[35–49] address different aspects of the problem providing varying results on the coacervate phase behavior.

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The details of these approaches are quite rich as an inevitable consequence of multiple variables in the system. The current status of these attempts is recently reviewed[50–52] with exhaustive references therein. It has been pointed out that the contributions from the excluded volume effect and corrections to the Debye-Hückel formalism are significant in improving the VOT.

Briefly, the main approaches were based on density functional theory, field theory, liquid state theory, and simulations. Using a density functional theory and imposing globule-like structures to polyelectrolyte complexes, Borue and Erukhimovich[35] computed the phase diagram without counterions. The role of stoichiometry of polyanions and polycations was considered by Zhang and Shklovskii.[36] Counterions are not considered in this study as well. In the approach with the field theory, the use of the random phase approximation (RPA) has been the standard practice. The roles of charge density,[37] impenetrability of ions and the Flory-Huggins χ parameter,[38] ion-pair formation,[39] and the fractal dimension of polymers[40] were addressed, within the framework of RPA. Using the field-theory based simulations (FTS), Fredrickson and collaborators[41–43] have computed phase diagrams of diblock polyelectrolytes and diblock polyampholytes, without the consideration of the counterion entropy. Interfacial tension between a coacervate phase and its supernatant was also calculated using the scheme of FTS[44] with counterions and salt, as a complement to combining VOT and Cahn-Hilliard theory.[45] Using a liquid state theory and a suitable closure approximation, Perry and Sing have constructed a phase diagram in qualitative agreement with their key experimental observations, with the conclusion that the effects of excluded volume and chain connectivity cancel each other, thus making the VOT useful.[46] This conclusion was further supported by Monte Carlo simulations.[47] There have also been attempts to consider the role of entropy associated with the counterion release,[53] based on ionization equilibria of charged monomers and water reorganization.[34,48,49,56] Salehi *et al.* and Lytle *et al.* have extended the VOT by accounting for the entropy associated with counterion adsorption equilibria and pairing of oppositely charged monomers. Based on experiments, Fu and Schlenoff[56] and Tirrell and co-workers[22–27] have emphasized the significance of the role of entropy due to the counterion release mechanism[53] in coacervation.

As evident from the above summary, the theoretical basis of coacervate phase behavior is far from a full understanding. While fluctuations in polyelectrolyte conformations and in the composition of solutions can contribute to the free energy of the system as in the field-theory based theories and all-atom simulations, it is instructive[52] to explicitly consider even the mean field theory of VOT and address rectification of the main objectionable approximations of VOT (in the absence of fluctuation effects). According to VOT, the free energy density f_{VO} of a symmetric solution (polycations and polyanions are equal in number and length) of polycations, polyanions, salts, and solvent is

$$f_{VO} = \frac{\phi_p}{N} \ln \frac{\phi_p}{2} + \phi_s \ln \frac{\phi_s}{2} + \phi_0 \ln \phi_0 - \frac{(\kappa_{VO}\ell)^3}{12\pi}, \quad (1)$$

where N is the number of Kuhn segments per chain and ϕ_p, ϕ_s, and ϕ_0 are the volume fractions of polyelectrolytes (both polycations and polyanions), added salt ions, and solvent molecules, respectively. In writing this expression, the volumes of polymer segments, fully dissociated salt ions, and solvent molecules are taken to be the cube of the Kuhn length ℓ. The inverse Debye screening length κ_{VO} in VO theory is

$$\kappa_{VO}^2 = \frac{4\pi\ell_B}{\ell^3}(\phi_s + \sigma\phi_p), \quad (2)$$

where ℓ_B is the Bjerrum length, ℓ is Kuhn segment length, and σ is the chain backbone charge density, which is the total charge number divided by N. The Bjerrum length ℓ_B is defined as

$$\ell_B = \frac{e^2}{4\pi\epsilon_0\epsilon k_B T}, \quad (3)$$

where e is the electronic charge, ϵ_0 is the permittivity of vacuum, ϵ is the dielectric constant of the medium, and $k_B T$ is the Boltzmann constant times the absolute temperature. In view of the temperature dependence of ℓ_B, it is convenient to define the reduced temperature t as

$$t = \frac{\ell}{4\pi\ell_B}. \quad (4)$$

The reduced temperature is converted to the absolute temperature using a choice of the Kuhn length ℓ and the dielectric constant ϵ.

The first three terms on the right hand side of Eq. (1) are from the translational entropy of the chains, added salt ions, and the solvent molecules. The last term is the Debye-Hückel electrostatic correlation free energy with the unphysical assumption that all polyelectrolyte chains are completely decomposed into their charged monomers. Even with this drastic assumption of neglect of charge connectivity, the factor σ in Eq. (1) must be σ^2 according to the Debye-Hückel theory.[52] For large values of N, the first term on the right hand side of Eq. (1) is negligible in comparison with the other terms. The phase behavior is generated by the VOT from the combination of essentially the entropy of salt ions and solvent molecules and electrostatic correlations among fully decomposed polymer moieties and salt ions. The VOT does not consider counterions from the polycations and polyanions.

Since the formulation of the VOT sixty years ago, the field of charged macromolecules has progressed significantly and it is straight forward to derive the correct free energy density at the mean field level of VOT. For a symmetric solution of polycations and polyanions (each polyelectrolyte with n chains and each chain with N segments) with uniform degree of dissociation α into their counterions, the combination of the Flory-Huggins theory (for the entropy of mixing of polycations, polyanions, counterions, salt ions, and solvent molecules) and the Debye-Hückel theory (for the electrostatic correlation among the various counterions and salt ions but not polymer segments) gives the free energy density f as[52]

$$f = (\frac{1}{N} + \alpha)\phi_p \ln \frac{\phi_p}{2} + \phi_s \ln \frac{\phi_s}{2} + \phi_0 \ln \phi_0 - \frac{(\kappa\ell)^3}{12\pi}, \quad (5)$$

where

$$\kappa^2 = \frac{4\pi\ell_B}{\ell^3}(\phi_s + \alpha\phi_p). \tag{6}$$

As in Eq. (1), ϕ_p, ϕ_s, and ϕ_0 are the volume fractions of the total polymer (both polycations and polyanions), salt ions, and solvent molecules, respectively.

The immiscibility of the polyelectrolyte backbone with water can be modeled by using the Flory-Huggins χ parameter so that both Eqs. (1) and (5) have an additional term

$$f_H = \chi\phi_p\phi_0, \tag{7}$$

and hence Eqs. (1) and (5) become

$$f_{VO} = \frac{\phi_p}{N}\ln\frac{\phi_p}{2} + \phi_s\ln\frac{\phi_s}{2} + \phi_0\ln\phi_0 - \frac{(\kappa_{VO}\ell)^3}{12\pi} + \chi\phi_p\phi_0 \tag{8}$$

and

$$f = (\frac{1}{N} + \alpha)\phi_p\ln\frac{\phi_p}{2} + \phi_s\ln\frac{\phi_s}{2} + \phi_0\ln\phi_0 - \frac{(\kappa\ell)^3}{12\pi} + \chi\phi_p\phi_0. \tag{9a}$$

If counterions are treated to be indistinguishable to the salt ions, Eq. (9a) becomes

$$f = \frac{1}{N}\phi_p\ln\frac{\phi_p}{2} + \phi_s'\ln\frac{\phi_s'}{2} + \phi_0\ln\phi_0 - \frac{(\kappa\ell)^3}{12\pi} + \chi\phi_p\phi_0, \tag{9b}$$

where ϕ_s' refers to the volume fraction of all the ions including salt ions and counterions and has a lower bound given by $\phi_s' \geq \alpha\phi_p$.

Within the Flory-Huggins theory, χ is inversely proportional to T, and since T appears in ℓ_B as well, both χ and ℓ_B are written in terms of a single reduced temperature variable t as[53]

$$\frac{\ell_B}{\ell} \equiv \frac{1}{4\pi t} \quad \text{and} \quad \chi \equiv \frac{a_\chi}{20\pi t}. \tag{10}$$

If χ were to be taken as $\Theta/2T$, where Θ is the Flory ideal temperature at which $\chi = 1/2$, the parameter a_χ $(=10\pi\epsilon_0\epsilon k_B\Theta\ell/e^2)$ reflects the magnitude of the Θ temperature and is a measure of the hydrophobic interaction between the polymers and the solvent.

There are two important aspects by which Eqs. (5) and (9) differ from Eqs. (1) and (8) of VOT. First, the contribution arising from the counterion entropy, $\alpha\phi_p\ln(\phi_p/2)$, is quite significant in comparison with that from the translational entropy of chains, $(1/N)\phi_p\ln(\phi_p/2)$, as $\alpha > 1/N$. Hence suppressing the $\alpha\phi_p\ln(\phi_p/2)$ term as done in the VOT is a serious ad hoc assumption. Second, the inverse Debye screening length κ in the correct formulation arises from the counterions and salt ions, whereas in VOT, it is from decomposed polymer moieties and salt ions without any consideration of the counterions (although the mathematical expression is superficially the same).

As an illustration of whether such mean field descriptions can provide reasonable coacervate phase diagrams pertinent to experimental results, we now consider sample predictions from Eqs. (8) and (9b). The computed phase diagram from Eq. (8) (VOT) for N = 100, σ = 0.24, and χ = 0 is given in Fig. 1(a) at t = 0.0375. For the choice of ℓ = 0.33 nm

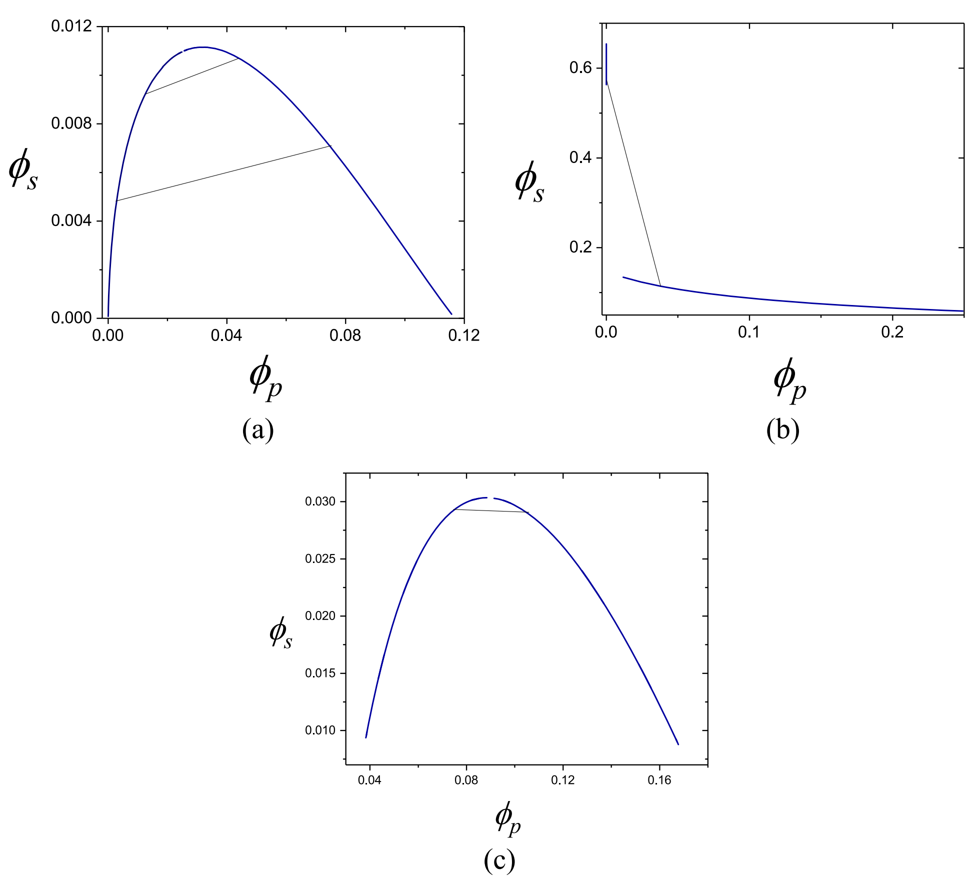

FIG. 1. Total ion concentration (ϕ_s) vs polymer concentration (ϕ_p) constructed by using (a) Eq. (8) with N = 100, σ = 0.24, and χ = 0 at t = 0.0375 (25 °C), (b) Eq. (9b) with N = 100, α = 0.24 and χ = 0 at t = 0.0375 (25 °C), and (c) Eq. (9b) with N = 100, α = 0.24, and χ = 0.62 at t = 0.051 (133 °C). For (a), since there are no counterions, total ions refer to salt ions only, while for (b) and (c), total ions refer to salt ions and counterions together. To calculate the real temperature, ℓ = 0.33 nm and ϵ = 80 are used. For salt free cases, the critical temperature for (b) −98 °C is found to be much lower than that for (a) 77 °C.

and $\epsilon = 80$, $t = 0.0375$ corresponds to 25 °C, as indicated in the figure. As recognized in the literature[14,20–26] this phase diagram is close to experimental findings. On the other hand, when the flawed assumptions of VOT are corrected, Eq. (9b) results in the phase diagram of Fig. 1(b) for the same values of parameters (by taking $\alpha = \sigma$) as in Fig. 1(a). It is evident from Figs. 1(a) and 1(b) that there is a significant discrepancy in the shape of the coexistence curve. Also, the critical temperature $t_c = 0.022$ (−98 °C) for the salt free case of Eq. (9b) is found to be much lower compared to that $t_c = 0.0440$ (77 °C) for the salt free case of Eq. (8). Furthermore, the critical temperature for Eq. (9b) increases with salt, while that for Eq. (8) decreases with salt concentration. The origin of these features lies in the incorrect definition of κ and imposed insignificant contribution from the translational entropy of polymer chains in the VOT. While the factor $(\alpha + 1/N)$ is of order unity in the corrected theory, the corresponding factor $(1/N)$ in the VOT is negligibly small. Therefore, in order to obtain coacervate phase separation at experimentally relevant temperatures, the hydrophobic interaction (χ) term must be significantly larger in order to promote phase separation. For the same values of the parameters used in Figs. 1(a) and 1(b), and $\chi = 0.62$, Eq. (9b) gives the phase diagram of Fig. 1(c) at an unphysically high temperature $t = 0.051$ (133 °C) [which is comparable to the shape in Fig. 1(a) at $t = 0.0375$ (25 °C)]. If the temperature is lower, then the shape of the phase diagram reverts to that of Fig. 1(b). The same shape of the phase diagram of Fig. 1(c) is obtained even at a temperature as high as $t = 0.070$ (284 °C), although a phase diagram of the shape of Fig. 1(c) is not obtained at room temperature for $\chi = 0.62$. Therefore, it is theoretically possible to obtain coacervate phase diagrams that look similar to experimental results by arbitrarily choosing a large value of χ.

The apparent success of VOT, despite its well-known approximations, and the above results of Fig. 1 suggest that any mechanism that can lead to an increase in the effective value of χ would result in coacervate phase diagrams pertinent to experimental results. Instead of empirically assigning a large value of χ, one of the present authors recently introduced a new model where additional polymer-polymer attractive interaction can strongly emerge from dipole-dipole interactions among partially complexed chains, as described below.

As recently argued,[52] the basic premise of the starting point in deriving the free energy of the homogeneous phase of a solution of polycations and polyanions in the above discussed theoretical procedures might be insufficient for adequately describing the system. Instead of the presumed random distribution of all chains (polycations and polyanions), the spontaneous ability of oppositely charged polyelectrolytes to form intermolecular complexes, such as ladders, branched architectures, and micelles, even in the homogeneous phase away from the domain of coacervate phase separation, must be accounted for. A complex between two oppositely charged polyelectrolytes is in general a network of "multiblock" chains with positive charges, negative charges, and dipoles distributed in random sequences. When the concentration is not very low, many chains can aggregate into a network. In fact, the formation of polyelectrolyte complexes in solutions has been widely recognized over more than five decades.[5,17,18] If the concentration is low, ladder-like conformations are more prominent as originally envisaged by Michaels,[5] Kabanov *et al.*,[18] and others.[17] Therefore a solution consisting of oppositely charged polyelectrolytes is made of a population of diverse architectures even in the homogeneous phase. A complete treatment of statistical mechanics of this population is a nontrivial task. Nevertheless, one of the limits in this physical picture is the presence of ladders, which forms the basis of the present paper as the zeroth-order model to capture the role of dipole formation. A cartoon of an illustrative complex formed by two oppositely charged chains is portrayed in Fig. 2. The spontaneous formation of such complexes has two effects: (i) release of counterions from both polycations and polyanions and (ii) formation of trains of dipoles, with loops and tails of charge-bearing polycations and polyanions. The formation of blocks of dipoles reduces the electrostatic repulsion among otherwise un-complexed chains and effectively increases the value of χ due to dipole-dipole interactions which are attractive between polymer segments. While the release of counterions increases the entropy[54–56] and hence stabilizes the homogeneous phase, the dipole formation favors coacervate phase separation.

In general, as suggested in Ref. 52, the presence of branched architectures of complexes and micelles formed by polycations and polyanions in the homogeneous phase is expected to lead to a rich phase diagram involving both liquid-liquid coacervate phase separation and gel-like coacervate phase. In an effort to discern whether this new model has any potential to capture the basic qualitative aspects of coacervate phenomenon, we address here the computation of phase diagrams based on the simple model introduced in Ref. 52. In this model, only pairwise ladder-like complexation, without multiple trains of dipoles, is allowed in the homogeneous phase. The attractive dipole-dipole

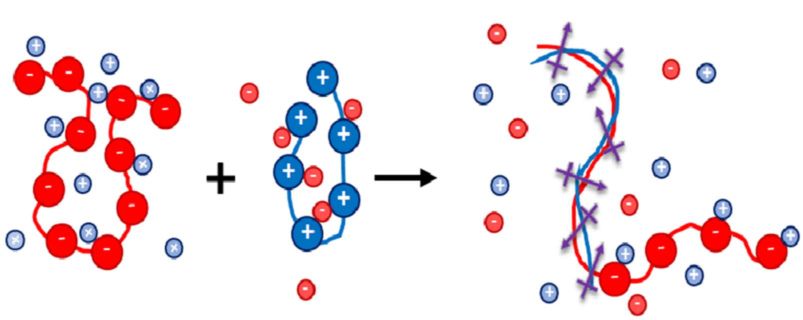

FIG. 2. Cartoon of pairwise complexation of two oppositely charged polyelectrolytes, driven by the release of counterions and formation of trains of dipoles. At higher polyelectrolyte concentrations, more complex tree-like structures can form in the homogeneous phase.

and dipole-charge interactions and the intrinsic hydrophobicity bring the pairwise complexes together to promote phase separation, which is countered by the charge-charge repulsive interactions among the unpaired blocks of the chains and counterion entropy. The phase diagrams calculated with even this simple model capture qualitatively the key experimental observations such as the slope of the tie lines of coexistence curves and the effects of salt concentration, while making experimentally testable predictions.

The rest of the paper is organized as follows. The model and theory are introduced in Sec. II. The computed results are discussed in Sec. III followed by Sec. IV.

II. MODEL AND THEORY

Oppositely charged polyelectrolytes in aqueous solutions spontaneously form intermolecular complexes, such as ladders, branched architectures, and micelles, even in the homogeneous phase away from the domain of coacervate phase separation. Such complexes are in general networks of "multiblock" chains with positive charges, negative charges, and dipoles distributed in certain sequences. When the polyelectrolyte concentration is not very low, many chains can aggregate into a network, as shown in Fig. 24 of Ref. 52. If the polyelectrolyte concentration is low, ladder-like conformations are more prominent as originally envisaged by Michaels,[5] Kabanov *et al.*,[18] and others.[17] Therefore a solution consisting of oppositely charged polyelectrolytes is made of a population of diverse architectures even in the homogeneous phase. A complete treatment of statistical mechanics of this population is a nontrivial task. Nevertheless, one of the limits in this physical picture is the presence of ladders, which forms the model of the present paper as the zeroth-order model to capture the role of dipole formation (Fig. 2).

We consider a polyelectrolyte solution of volume V containing n_1 polycations of N_1 Kuhn segments, n_2 polyanions of N_2 Kuhn segments, counterions, salt ions, and n_0 solvent molecules. Let α be the degree of ionization for both the polycations and polyanions. Driven by the entropy gain of counterion release, we assume that each available polycation spontaneously binds with one available polyanion partially neutralizing their backbone charges. We consider each pair complex as a chemically distinct species from the uncomplexed polyelectrolytes. This species is a chain of a block of dipoles of varying orientation and blocks of polyelectrolyte segments, and because they are chemically distinct, we introduce them as a third polyelectrolyte component whose total number equals either n_1 or n_2, whichever is smaller. This third component replaces one of the other two components by still maintaining the same number of thermodynamic components of the system. There is then an excess of unpaired polyelectrolyte chains equal to the difference between n_1 and n_2. Different cases can be considered depending upon the relative magnitudes of n_1, n_2, N_1, and N_2. We consider a specific case when the polyanion length is longer than or equal to that of the polycation (Fig. 2), and also the number of polyanions is greater than or equal to that of polycations, that is, $N_1 \leq N_2$ and $n_1 \leq n_2$. In addition, let the shorter chain occupy the contour of the longer chain from s_1 to $s_1 + N_1$, as depicted in Fig. 2. Thus the system is composed of n_1 pair complexes, $n_{2'} = n_2 - n_1$ excess polyanions which are unpaired, counterions, salt ions, and solvent molecules. For simplicity, the valencies of all charged species are taken to be 1. We assume that the counterions previously coupled with the paired portions of the complexes are liberated as free ions, while the unpaired portions of the complexes and the excess polyanions release their counterions with a degree of ionization α. In total, there are $n_{c1} = n_1 N_1$ negatively charged counterions and $n_{c2} = (1-\alpha)n_1 N_1 + \alpha n_2 N_2$ positively charged counterions. In addition, there are n_+ cations and n_- anions from the fully dissociated additional salt. To summarize, the solution now comprises of n_1 pair complexes, $n_2 - n_1$ unpaired polyanions, n_{c1} negatively charged counterions, n_{c2} positively charged counterions, n_+ cations, n_- anions, and n_0 solvent molecules. All ions, solvent molecules, and polyelectrolyte segments are considered to occupy a lattice site of length ℓ and the system is incompressible. The dielectric constant ϵ is assumed to be constant throughout the solution.

The Helmholtz free energy F of the system is given by

$$e^{-\frac{F}{k_BT}} = \frac{1}{n_1!(n_2-n_1)!n_{c1}!n_{c2}!n_0!\prod_\gamma n_\gamma!}\int\prod_{\alpha=1}^{n_1}\mathcal{D}[\mathbf{R}_\alpha(s_\alpha)] \times \int\prod_{\beta=1}^{n_2}\mathcal{D}\big[\mathbf{R}_\beta\big(s_\beta\big)\big]\prod_i^{n_{c1}+n_{c2}+n_0+\sum_\gamma n_\gamma} d\mathbf{r}_i \times \int_0^{N_2-N_1} ds_1 \exp\left[-(H_0 + U_{pp,el} + U_{pp,ex} + U_{p0} + U_{00} + U_{pi} + U_{ii})\right]. \tag{11}$$

Here $\mathcal{D}[\mathbf{R}_\alpha(s_\alpha)]$ denotes the functional derivative of the arc length variable s_α of chain α $(0 \leq s_\alpha \leq N_1)$, and similarly $\mathcal{D}[\mathbf{R}_\beta(s_\beta)]$ corresponds to the β chains $(0 \leq s_\beta \leq N_2)$. H_0 represents the connectivity of chains. The electrostatic part and the excluded volume part of polymer-polymer interactions are represented by $U_{pp,el}$ and $U_{pp,ex}$, respectively. U_{ps}, U_{ss}, U_{pi}, and U_{ii} represent interactions among polymer-solvent, solvent-solvent, polymer-ion, and ion-ion, respectively. These are modeled as follows.

The chain connectivity term H_0 is given by the Edwards Hamiltonian,

$$H_0 = \frac{3}{2\ell^2}\sum_{\alpha=1}^{n_1}\int_0^{N_1} ds_\alpha\left(\frac{\partial\mathbf{R}_\alpha(s_\alpha)}{\partial s_\alpha}\right)^2 + \frac{3}{2\ell^2}\sum_{\beta=1}^{n_2}\int_0^{N_2} ds_\beta\left(\frac{\partial\mathbf{R}_\beta(s_\beta)}{\partial s_\beta}\right)^2. \tag{12}$$

The electrostatic contribution of polymer-polymer interaction is given by

$$
\begin{aligned}
U_{pp,el} = &\frac{1}{2}\sum_{\alpha=1}^{n_1}\sum_{\alpha'=1}^{n_1}\{\int_0^{s_1} ds_\alpha[\int_0^{s_1} ds_{\alpha'}u_{cc}^0(\mathbf{R}_{\alpha\alpha'}) + \int_{s_1}^{s_1+N_1} ds_{\alpha'}u_{cd}^0(\mathbf{R}_{\alpha\alpha'}) + \int_{s_1+N_1}^{N_2} ds_{\alpha'}u_{cc}^0(\mathbf{R}_{\alpha\alpha'})] \\
&+ \int_{s_1}^{s_1+N_1} ds_\alpha[\int_0^{s_1} ds_{\alpha'}u_{cd}^0(\mathbf{R}_{\alpha\alpha'}) + \int_{s_1}^{s_1+N_1} ds_{\alpha'}u_{dd}^0(\mathbf{R}_{\alpha\alpha'}) + \int_{s_1+N_1}^{N_2} ds_{\alpha'}u_{cd}^0(\mathbf{R}_{\alpha\alpha'})] \\
&+ \int_{s_1+N_1}^{N_2} ds_\alpha[\int_0^{s_1} ds_{\alpha'}u_{cc}^0(\mathbf{R}_{\alpha\alpha'}) + \int_{s_1}^{s_1+N_1} ds_{\alpha'}u_{cd}^0(\mathbf{R}_{\alpha\alpha'}) + \int_{s_1+N_1}^{N_2} ds_{\alpha'}u_{cc}^0(\mathbf{R}_{\alpha\alpha'})]\} \\
&+ \frac{1}{2}\sum_{\beta=1}^{n_2-n_1}\sum_{\beta'=1}^{n_2-n_1}\int_0^{N_2} ds_\beta \int_0^{N_2} ds_{\beta'}u_{cc}^0(\mathbf{R}_{\beta\beta'}) + \sum_{\alpha=1}^{n_1}\sum_{\beta=1}^{n_2-n_1}\int_0^{N_2} ds_\beta[\int_0^{s_1} ds_\alpha u_{cc}^0(R_{\alpha\beta}) \\
&+ \int_{s1}^{s_1+N_1} ds_\alpha u_{cd}^0(\mathbf{R}_{\alpha\beta}) + \int_{s_1+N_1}^{N_2} ds_\alpha u_{cc}^0(\mathbf{R}_{\alpha\beta})],
\end{aligned} \tag{13}
$$

where $\mathbf{R}_{\alpha\alpha'}$, $\mathbf{R}_{\beta\beta'}$, and $\mathbf{R}_{\alpha\beta}$ denote $\mathbf{R}_\alpha(s_\alpha) - \mathbf{R}_{\alpha'}(s_{\alpha'})$, $\mathbf{R}_\beta(s_\beta) - \mathbf{R}_{\beta'}(s_{\beta'})$, and $\mathbf{R}_\alpha(s_\alpha) - \mathbf{R}_\beta(s_\beta)$, respectively. In the above equation, u_{cc}^0, u_{cd}^0, and u_{dd}^0 are the bare charge-charge, charge-dipole, and dipole-dipole interaction energies for polymer segments,

$$
\begin{aligned}
u_{cc}^0(\mathbf{r}) &= \frac{\alpha^2\ell_B}{r}, \quad u_{cd}^0(\mathbf{r}) = \frac{\alpha\ell_B}{r^3}\mathbf{p}\cdot\mathbf{r}, \\
u_{dd}^0(\mathbf{r}) &= \frac{\ell_B}{r^3}[\mathbf{p}_1\cdot\mathbf{p}_2 - 3(\mathbf{n}\cdot\mathbf{p}_1)(\mathbf{n}\cdot\mathbf{p}_2)].
\end{aligned} \tag{14}
$$

Here $\mathbf{p}$, $\mathbf{p}_1$, and $\mathbf{p}_2$ denote the dipole moments formed by the complexing charged monomers, $\mathbf{n}$ is the unit vector in the direction of $\mathbf{r}$, and r is the magnitude of $\mathbf{r}$.

The contribution from the excluded volume interactions is given by

$$
\begin{aligned}
U_{pp,ex} = &\frac{1}{2}\sum_{\alpha=1}^{n_1}\sum_{\alpha'=1}^{n_1}\int_0^{N_1} ds_\alpha \int_0^{N_1} ds_{\alpha'} w_{pp}\ell^3\delta(\mathbf{R}_\alpha(s) - \mathbf{R}_{\alpha'}(s)) \\
&+ \frac{1}{2}\sum_{\beta=1}^{n_2}\sum_{\beta'=1}^{n_2}\int_0^{N_2} ds_\beta \int_0^{N_2} ds_{\beta'} w_{pp}\ell^3\delta(\mathbf{R}_\beta(s) - \mathbf{R}_{\beta'}(s)) \\
&+ \sum_{\alpha=1}^{n_1}\sum_{\beta=1}^{n_2}\int_0^{N_1} ds_\alpha \int_0^{N_2} ds_\beta w_{pp}\ell^3\delta(\mathbf{R}_\alpha(s) - \mathbf{R}_\beta(s)),
\end{aligned} \tag{15}
$$

where the excluded volume parameter w_{pp} is a pseudo potential depicting the binary cluster integral corresponding to the pair-wise short-ranged hydrophobic segment-segment interaction. Analogously, U_{ps} and U_{ss} in Eq. (11) for the short-ranged interactions between the polymer segments and solvent molecules, and between the solvent molecules, are given by

$$
\begin{aligned}
U_{p0} = &\sum_{\alpha=1}^{n_1}\int_0^{N_1} ds_\alpha \sum_{i=1}^{n_0} w_{p0}\ell^3\delta(\mathbf{R}_\alpha(s) - \mathbf{r}_i) \\
&+ \sum_{\beta=1}^{n_2}\int_0^{N_2} ds_\beta \sum_{i=1}^{n_0} w_{p0}\ell^3\delta(\mathbf{R}_\beta(s) - \mathbf{r}_i)
\end{aligned} \tag{16}
$$

and

$$
U_{00} = \frac{1}{2}\sum_{i=1}^{n_0}\sum_{j=1}^{n_0} w_{00}\ell^3\delta(\mathbf{r}_i - \mathbf{r}_j). \tag{17}
$$

The interaction terms involving the dissociated ions are given by

$$
\begin{aligned}
U_{pi} = &\sum_{\alpha=1}^{n_1}\sum_{i=1}^{n_{c1}+n_{c2}+\sum_\gamma n_\gamma}\Bigg[\int_0^{s_1} ds_\alpha u_{ci}^0(\mathbf{R}_\alpha(s_\alpha) - \mathbf{r}_i) \\
&+ \int_{s_1}^{s_1+N_1} ds_\alpha u_{di}^0(\mathbf{R}_\alpha(s_\alpha) - \mathbf{r}_i) \\
&+ \int_{s_1+N_1}^{N_2} ds_\alpha u_{ci}^0(\mathbf{R}_\alpha(s_\alpha) - \mathbf{r}_i)\Bigg] \\
&+ \sum_{\beta=1}^{n_2-n_1}\sum_{i=1}^{n_{c1}+n_{c2}+\sum_\gamma n_\gamma}\int_0^{N_2} ds_\beta u_{ci}^0(\mathbf{R}_\beta(s_\beta) - \mathbf{r}_i)
\end{aligned} \tag{18}
$$

and

$$
U_{ii} = \frac{1}{2}\sum_{i=1}^{n_{c1}+n_{c2}+\sum_\gamma n_\gamma}\sum_{j=1}^{n_{c1}+n_{c2}+\sum_\gamma n_\gamma} u_{ii}^0(\mathbf{r}_i - \mathbf{r}_j), \tag{19}
$$

where

$$
u_{ci}^0(\mathbf{r}) = \frac{\alpha\ell_B}{r}, \quad u_{di}^0(\mathbf{r}) = \frac{\ell_B}{r^3}\mathbf{p}\cdot\mathbf{r}, \quad u_{ii}^0(\mathbf{r}) = \frac{\ell_B}{r}. \tag{20}
$$

Integration over the coordinates of counterions and salt ions in Eq. (11), and in combination with the corresponding factorial terms, decouples polymers and the ionic background fluid.[57] Now, the bare polymer-polymer interactions u_{cc}^0, u_{cd}^0, and u_{dd}^0 of Eq. (14) are converted into their respective screened interactions u_{cc}, u_{cd}, and u_{dd}, as given below. The free energy can be written as the sum of contributions from the polymer F_p and the background fluid F_b,

$$
F = F_p + F_b. \tag{21}
$$

The free energy of the background fluid consists of the entropy of mixing terms and the charge fluctuation term as[57]

$$F_b = F_{b0} + F_{fl,i}, \tag{22}$$

where

$$\frac{F_{b0}\ell^3}{k_BTV} = \phi_{c1}\ln\phi_{c1} + \phi_{c2}\ln\phi_{c2} + \phi_+\ln\phi_+ + \phi_-\ln\phi_- \tag{23}$$

and

$$\frac{F_{fl,i}\ell^3}{k_BTV} = -\frac{1}{4\pi}[\ln(1+\kappa\ell) - \kappa\ell + \frac{1}{2}\kappa^2\ell^2], \tag{24}$$

with the inverse Debye length κ given by

$$\kappa^2\ell^2 = \frac{4\pi\ell_B}{\ell}(\phi_{c1} + \phi_{c2} + \phi_+ + \phi_-), \tag{25}$$

where $\phi_{c1} = n_{c1}\ell^3/V$, $\phi_{c2} = n_{c2}\ell^3/V$, $\phi_+ = n_+\ell^3/V$, and $\phi_- = n_-\ell^3/V$. For $N_2 > N_1$, the entropy of aligning the polycation and polyanion is $n_1 \ln(N_2 - N_1)$, which leads to only a linear term in ϕ_1 in Eq. (23). Therefore, the entropic contribution arising from the number of different ways of lining up the polycation and polyanion does not affect the calculation of the phase diagrams.

The polymer contribution F_p is obtained by performing the remaining integrals in Eq. (11) using the screened expressions for the charge-charge, charge-dipole, and dipole-dipole interactions among polymer segments

$$u_{cc}(\mathbf{r}) = \frac{\alpha^2\ell_B}{r}e^{-\kappa r},$$

$$u_{cd}(\mathbf{r}) = \frac{\alpha\ell_B}{r^3}(\mathbf{p}\cdot\mathbf{r})e^{-\kappa r}(1+\kappa r),$$

$$u_{dd}(\mathbf{r}) = \frac{\ell_B}{r^3}e^{-\kappa r}[(1+\kappa r)\mathbf{p}_1\cdot\mathbf{p}_2 - (3+3\kappa r+\kappa^2r^2)(\mathbf{n}\cdot\mathbf{p}_1)(\mathbf{n}\cdot\mathbf{p}_2)]. \tag{26}$$

Since we focus only on the mean field level by relegating the role of conformational fluctuations and micellization to a future work, we write the electrostatic and excluded volume interactions as two-body pseudo potentials. For example, $U_{pp,ex}$, U_{ps}, and U_{ss} are

$$\frac{U_{pp,ex}\ell^3}{V} = \frac{1}{2}w_{pp}(\phi_1+\phi_2)^2, \quad \frac{U_{p0}\ell^3}{V} = w_{p0}(\phi_1+\phi_2)\phi_0,$$

and

$$\frac{U_{00}\ell^3}{V} = \frac{1}{2}w_{00}\phi_0^2, \tag{27}$$

where $\phi_1 = n_1N_1\ell^3/V$, $\phi_2 = n_2N_2\ell^3/V$, and $\phi_0 = n_0\ell^3/V$. The electrostatic part of the polymer-polymer interaction is obtained from Eq. (13) [by using the screened interaction terms of Eq. (26) instead of the bare terms of Eq. (20)] as

$$\frac{U_{pp,el}\ell^3}{V} = \frac{1}{2}v_{cc}(\phi_2-\phi_1)^2 + \frac{1}{2}v_{dd}\phi_1^2 + v_{cd}\phi_1(\phi_2-\phi_1), \tag{28}$$

where the pseudo potentials $v_{\alpha\beta}$ ($\alpha = c, d$ and $\beta = c, d$)are defined as

$$v_{\alpha\beta} = \frac{1}{\ell^3}\int d\mathbf{r}[1 - e^{-u_{\alpha\beta}(\mathbf{r})}]. \tag{29}$$

Noting that $u_{cc}(\mathbf{r})$ approaches $(4\pi\alpha^2\ell_B/\kappa^2)\delta(\mathbf{r})$ for higher concentrations of counterions and salt ions, and performing the high-temperature expansion, and keeping only the leading term, v_{cc} follows from Eqs. (26) and (29) as

$$v_{cc} = \frac{4\pi\alpha^2\ell_B}{\kappa^2\ell^3}. \tag{30}$$

For the dipolar terms v_{cd} and v_{dd}, we first perform the angular averages on $u_{cd}(\mathbf{r})$ and $u_{dd}(\mathbf{r})$ to obtain $\langle u_{cd}\rangle$ and $\langle u_{dd}\rangle$ as defined by

$$e^{-\langle u_{cd}\rangle} = \frac{\int d\Omega e^{-u_{cd}(\mathbf{r})}}{\int d\Omega} \quad \text{and} \quad e^{-\langle u_{dd}\rangle} = \frac{\int d\Omega\int d\Omega' e^{-u_{dd}(\mathbf{r})}}{\int d\Omega\int d\Omega'}, \tag{31}$$

where Ω and Ω' denote the solid angles for the dipole orientations. In the high-temperature limit, $\langle u_{cd}\rangle$ and $\langle u_{dd}\rangle$ are obtained as

$$\langle u_{cd}\rangle = -\frac{1}{6}\frac{\alpha^2\ell_B^2p^2}{r^4}e^{-2\kappa r}(1+\kappa r)^2, \tag{32}$$

$$\langle u_{dd}\rangle = -\frac{1}{3}\frac{\ell_B^2p_1^2p_2^2}{r^6}e^{-2\kappa r}[1 + 2\kappa r + \frac{5}{3}(\kappa r)^2 + \frac{2}{3}(\kappa r)^3 + \frac{1}{6}(\kappa r)^4]. \tag{33}$$

Combining Eqs. (29), (32), and (33), the pseudo potentials v_{cd} and v_{dd} become (in the high temperature expansion limit and implementing the shortest separation distance as ℓ)

$$v_{cd} = -\frac{\pi}{3}\frac{\alpha^2\ell_B^2p^2}{\ell^4}e^{-2\kappa\ell}(2+\kappa\ell) \tag{34}$$

and

$$v_{dd} = -\frac{\pi}{9}\frac{\ell_B^2p_1^2p_2^2}{\ell^6}e^{-2\kappa\ell}[4 + 8\kappa\ell + 4(\kappa\ell)^2 + (\kappa\ell)^3]. \tag{35}$$

The combinatorial factors $n_1!(n_2-n_1)!$ along with the factor $V^{n_1}V^{n_2-n_1}$ ($=V^{n_2}$) associated with the translation of centers of mass of all chains in the system lead to the free energy contribution F_{Sp} as

$$\frac{F_{Sp}}{k_BT} = n_1\ln\frac{n_1(N_1+N_2)\ell^3}{V} + (n_2-n_1)\ln\frac{(n_2-n_1)N_2\ell^3}{V}, \tag{36}$$

where the irrelevant constant terms and terms linear in n_1 and n_2 are ignored. The corresponding free energy density $f_{Sp} \equiv F_{Sp}\ell^3/k_BTV$ is

$$f_{Sp} = \frac{\phi_1}{N_1}\ln[\phi_1(1+\frac{N_2}{N_1})] + (\frac{\phi_2}{N_2} - \frac{\phi_1}{N_1})\ln(\phi_2 - \phi_1\frac{N_2}{N_1}). \tag{37}$$

Similarly $V^{n_0}/n_0!$ in Eq. (11) results in the translational entropy contribution from the solvent to the free energy density as

$$f_{S0} = \phi_0\ln\phi_0. \tag{38}$$

Combining Eqs. (22)–(24), (27), (28), and (37), the total free energy density f becomes

$$f = f_{Sp} + f_{Si} + f_{S0} + f_{el} + f_{ex} + f_{fl,i} + f_{fl,p}, \tag{39}$$

where f_{Sp}, f_{S0}, and $f_{fl,i}$ are given by Eqs. (27), (28), and (37), and

$$f_{Si} = \phi_1\ln\phi_1 + [(1-\alpha)\phi_1 + \alpha\phi_2]\ln[(1-\alpha)\phi_1 + \alpha\phi_2] + \phi_+\ln\phi_+ + \phi_-\ln\phi_-, \tag{40}$$

$$f_{el} = \frac{1}{2}v_{cc}(\phi_2 - \phi_1)^2 + \frac{1}{2}v_{dd}\phi_1^2 + v_{cd}\phi_1(\phi_2 - \phi_1), \tag{41}$$

and

$$f_{ex} = \frac{1}{2}w_{pp}(\phi_1 + \phi_2)^2 + w_{p0}(\phi_1 + \phi_2)\phi_0 + \frac{1}{2}w_{00}\phi_0^2. \tag{42}$$

As already pointed out, $f_{fl,p}$ corresponding to the conformational fluctuations is ignored in the present paper.

If the counterions and the salt ions are indistinguishable, f_{Si} of Eq. (40) must be replaced by

$$f_{Si} = (\phi_1 + \phi_-)\ln(\phi_1 + \phi_-) + [(1-\alpha)\phi_1 + \alpha\phi_2 + \phi_+] \times \ln[(1-\alpha)\phi_1 + \alpha\phi_2 + \phi_+], \tag{43}$$

and the other terms in Eq. (39) remain unchanged.

In summary, the free energy density for the present model system in the homogeneous phase becomes (with $\phi_+ = \phi_- = \phi_s/2$, where ϕ_s is the total concentration of cations and anions from the added salt)

$$f = f_{Sp} + f_{el} + f_{ex} + f_{Si} + f_{fl,i} + f_{S0}, \tag{44}$$

where

$$f_{Sp} = \frac{\phi_1}{N_1}\ln[\phi_1(1 + \frac{N_2}{N_1})] + (\frac{\phi_2}{N_2} - \frac{\phi_1}{N_1})\ln(\phi_2 - \phi_1\frac{N_2}{N_1}), \tag{45}$$

$$f_{el} = \frac{1}{2}v_{cc}(\phi_2 - \phi_1)^2 + \frac{1}{2}v_{dd}\phi_1^2 + v_{cd}\phi_1(\phi_2 - \phi_1), \tag{46}$$

and

$$f_{ex} = \frac{1}{2}w_{pp}(\phi_1 + \phi_2)^2 + w_{p0}(\phi_1 + \phi_2)\phi_0 + \frac{1}{2}w_{00}\phi_0^2. \tag{47}$$

If the counterions are distinguishable from the salt ions,

$$f_{Si} = \phi_1\ln\phi_1 + [(1-\alpha)\phi_1 + \alpha\phi_2] \times\ln[(1-\alpha)\phi_1 + \alpha\phi_2] + \phi_s\ln\frac{\phi_s}{2}. \tag{48}$$

On the other hand, if the counterions and salt ions are indistinguishable,

$$f_{Si} = (\phi_1 + \frac{\phi_s}{2})\ln(\phi_1 + \frac{\phi_s}{2}) + [(1-\alpha)\phi_1 + \alpha\phi_2 + \frac{\phi_s}{2}] \times \ln[(1-\alpha)\phi_1 + \alpha\phi_2 + \frac{\phi_s}{2}]. \tag{49}$$

The term from the electrostatic correlation among the small ions is

$$f_{fl,i} = -\frac{1}{4\pi}[\ln(1 + \kappa\ell) - \kappa\ell + \frac{1}{2}\kappa^2\ell^2], \tag{50}$$

with the inverse Debye length κ given by

$$\kappa^2\ell^2 = \frac{4\pi\ell_B}{\ell}(2\phi_1 + \alpha(\phi_2 - \phi_1) + \phi_s), \tag{51}$$

and the entropic contribution from the solvent is

$$f_{S0} = \phi_0\ln\phi_0. \tag{52}$$

The two variables ϕ_1 and ϕ_2 can be equivalently expressed in terms of the total polymer concentration ϕ_p and the excess polymer concentration ϕ_{ex} defined by

$$\phi_p = \phi_1 + \phi_2 \qquad \text{and} \qquad \phi_{ex} = \frac{(n_2 - n_1)N_2\ell^3}{V} = \phi_2 - \frac{1}{y}\phi_1, \tag{53}$$

where the polymer length asymmetry ratio y is defined as

$$y = \frac{N_1}{N_2}. \tag{54}$$

In terms of ϕ_p and ϕ_{ex}, ϕ_1 and ϕ_2 are given by

$$\phi_1 = (\frac{y}{1+y})(\phi_p - \phi_{ex}) \quad \text{and} \quad \phi_2 = (\frac{1}{1+y})\phi_p + (\frac{y}{1+y})\phi_{ex}. \tag{55}$$

Substitution of Eq. (55) into Eqs. (44)–(52) gives the free energy density of the homogeneous phase in terms of ϕ_p and ϕ_{ex}.

When phase separation takes place with two coexisting phases with $\phi_{ex} \neq 0$, there are 15 variables (the two sets of values of ϕ_1, ϕ_2, ϕ_{c1}, ϕ_{c2}, ϕ_+, ϕ_-, and ϕ_0 corresponding to the two phases and the volume fraction x of one phase) with 9 constraints (two incompressibility conditions, one electroneutrality condition, and six lever rules). Therefore the free energy density of the system needs to be minimized with six independent variables. In the simpler case with $\phi_{ex} = 0$, the minimization is done with five independent variables. After computing the volume fractions of the various components in the coexisting phases, the total polymer concentration ϕ_p and the total ion concentration ϕ_i ($=\phi_{c1} + \phi_{c2} + \phi_+ + \phi_-$) are used to construct the effective phase diagrams of ϕ_i versus ϕ_p plots.

Specifically for the general situation of $n_2 \neq n_1$ and the counterions being distinguishable from the salt ions, the various species are (1) n_1 complexed chains with total volume fraction $\phi_1(1 + N_2/N_1)$, where $\phi_1 = n_1N_1\ell^3/V$ is the polycation volume fraction; (2) $n_{2'} = n_2 - n_1$ un-complexed polyanions, each with N_2 segments so that the total volume fraction of this species is $\phi_2 - \phi_1(N_2/N_1) = \phi_{ex}$; (3) negatively charged counterions from the polycations with the total volume fraction $n_{c1}\ell^3/V = \phi_1$; (4) positively charged counterions from the polyanions with the total volume fraction $n_{c2}\ell^3/V = (1 - \alpha)\phi_1 + \alpha\phi_2$; (5) salt cations with volume fraction $\phi_s/2$; (6) salt anions with volume fraction $\phi_s/2$; and (7) n_0 solvent molecules with volume fraction ϕ_0. Thus there are 14 variables (7 for each of the two coexisting phases) and the additional variable x for the fraction of one of the phases. In determining the values of these 15 quantities, there are 9 constraints: two conditions for the incompressibility (one for each of the coexisting phases), one condition for the electroneutrality of a phase (which automatically ensures electroneutrality of the other phase), and six lever rules for any six components (and that for the seventh species being redundant due to the imposed incompressibility condition). The multi-dimensional nonlinear minimization is performed using the Simplex algorithm.[58] The Appendix provides details of the solved equations for the various cases of number of components and whether the ions are distinguishable or not.

III. RESULTS AND DISCUSSION

A wide range of phase behavior is explored by constructing phase diagrams with varying experimentally relevant variables: temperature, salt concentration, the polyelectrolyte chain length, the polycation-polyanion composition ratio, and solvent-polyelectrolyte interaction energy. In our calculations, we have taken α = 1/3 (the degree of ionization of uncomplexed part of a chain), ϵ = 80 (the dielectric constant of the solvent), and $\ell = p_1 = p_2 = p = 0.55$ nm. The relative strengths of the electrostatic interactions, which are measured by ℓ_B (Bjerrum length), and the polyelectrolyte-solvent interaction which is given by w_{p0}, determine the phase behavior of the system. Since we ignore the chemical mismatch interactions with salt ions and set the polyelectrolyte-polyelectrolyte and solvent-solvent interaction energies to zero, $w_{pp} = w_{00} = 0$, w_{p0} is equivalent to the Flory χ parameter. The real temperature (T) dependence of solvent-polyelectrolyte interaction strength w_{p0} can be written as $w_{p0} = \frac{\Theta}{2T}$, where Θ is the Flory theta temperature. This relation can be rewritten in terms of reduced temperature as $w_{p0} = \frac{a_\chi}{20\pi t}$, where $a_\chi = 10\pi\epsilon_0\epsilon k_B\Theta\ell/e^2$ is taken as the solvent-polyelectrolyte interaction parameter. For a_χ = 1.7, the theta temperature is 257 K for ℓ = 0.55 nm and ϵ = 80. Also, for a_χ = 1.7, ℓ = 0.55 nm, and ϵ = 80, w_{p0} (or the χ parameter) at room temperature is 0.434 which corresponds to a moderately hydrophobic polyelectrolyte.

To explore the interplay of hydrophobic and electrostatic dipolar interactions, we choose a_χ = 1.7, which corresponds to w_{p0} = 0.434 at room temperature (as an example). Effects of various experimental variables to the phase behavior are presented below. For such moderately hydrophobic systems, we consider two cases: (a) solutions with counterions and (b) solutions without counterions pertaining to the experiments when counterions are washed out first and salt ions are added later.

A. Solutions with counterions

1. Effect of temperature and salt

In order to study the effect of temperature and salt, we consider a symmetric solution where polycations and polyanions are of equal lengths ($N_1 = N_2 = 100$) and are in equal number ($n_1 = n_2$). Figure 3 illustrates the binodals for total ions versus polyelectrolyte concentrations at three different temperatures. Note that since there will always be a certain amount of counterions in the homogeneous phase, only those homogeneous phases with (ϕ_i, ϕ_p) lying at the left of the black dotted line passing through the origin are experimentally allowed unless the counterions are taken out before adding salt (as in Ref. 47, for example). The slope of the black dotted line is given by the degree of ionization of the polyelectrolyte chains.

Comparison of three curves in Fig. 3 shows that increasing temperature inhibits complex coacervation demonstrated by the narrower region of instability at higher temperature. For the set of parameters used in Fig. 3, coacervation no longer occurs once the temperature reaches t = 0.066 95 (47 °C). Also, at a given temperature, the solution becomes stable above a threshold salt concentration because of screening of the attractive dipolar interactions by the salt ions. The negative slopes of the tie lines demonstrate that salt preferentially partitions into the polyelectrolyte poor phase.

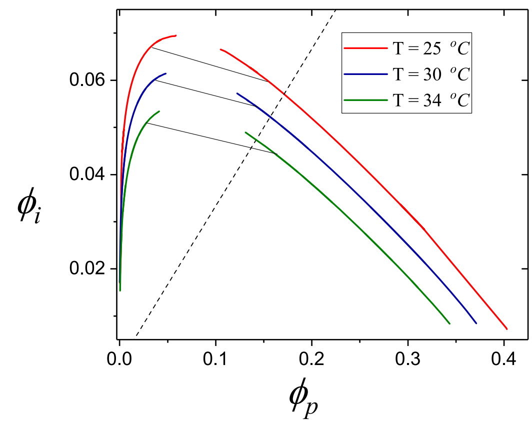

FIG. 3. Total ion (ϕ_i) vs polyelectrolyte (ϕ_p) concentration for a symmetric solution with $N_1 = N_2 = 100$, $n_1 = n_2$ for ϵ = 80, a_χ = 1.7, and $p = \ell$ = 0.55 nm. Salt ions and counterions are distinguishable. Curves represent binodals at t = 0.062 39 (25 °C) (red, outermost curve), t = 0.063 39 (30 °C) (blue, middle curve) and t = 0.064 39 (34 °C) (green, innermost curve) including tie lines (black). The black dotted line (representing the total concentration of dissociated counterions only, $\phi_i = \alpha\phi_p$) passing through the origin separates experimentally accessible homogeneous phases from those which are not accessible (explained in the text). Coacervation occurs only below 47 °C for this set of parameters.

This partitioning of salt arises from a combination of the electrostatic screening and excluded volume effect. The total free energy of the system can be reduced by expelling salt ions from the dipole rich phase, which strengthens the attractive dipolar interactions and hence reduces the electrostatic energy. Since the removal of salt ions from a particular phase is entropically unfavorable, the competition of two effects sets the tie line slope.

One important observation here is the enhancement of hydrophobicity by electrostatic dipolar interactions. Phase separation occurs even though the value of χ is not high enough for phase separation (χ = 0.434, 0.427, and 0.420 at 25 °C, 30 °C, and 34 °C, respectively). This is because the additional drive required is provided by dipolar attractions. The effects of salt as well as the slope of tie lines illustrated by Fig. 3 are in good qualitative agreement with experiments.[14,20–26,47] Our prediction on the temperature effect needs to be verified by future experiments.

2. Effect of polycation-polyanion composition asymmetry

The asymmetry in the polycation-polyanion content can be defined by a parameter A as $A = n_1N_1/n_2N_2$. First, we investigate the influence of the chain length asymmetry in the phase behavior by considering the case of $n_1 = n_2$, $N_1/N_2 = y$, and N_2 = 100. The phase diagrams in Fig. 4 are the binodals for different values of chain length asymmetry y. It shows how coacervation becomes less prevalent with decreasing symmetry, demonstrated by the shrinking region of phase instability. With decreasing the value of y, the area of the unstable region decreases. For the set of parameters used here, coacervation no longer occurs when the symmetry parameter y is below 0.86. The shrinking of the unstable region with decreasing y is due to the reduction of dipole-dipole attractions and an increase of charge-charge repulsions.

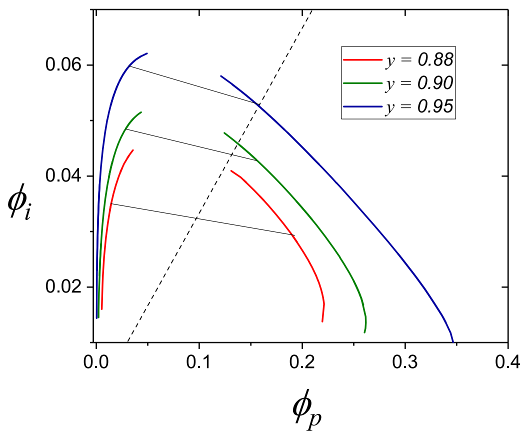

FIG. 4. Total ion (ϕ_i) vs polyelectrolyte (ϕ_p) concentration for an asymmetric solution with $n_1 = n_2$, $N_1/N_2 = y$, $N_2 = 100$ for $\epsilon = 80$, $a_\chi = 1.7$, and $p = \ell = 0.55$ nm at $t = 0.062\,39$ (25 °C). Salt ions and counterions are distinguishable. Curves represent binodals for different degrees of asymmetry given by $y = 0.95$ (blue, outermost curve), $y = 0.90$ (green, middle curve), and $y = 0.88$ (red, innermost curve). The black dotted line passing through the origin separates experimentally accessible homogeneous phases from those which are not accessible, as in Fig. 3.

Second, we take the case of $n_1 < n_2$ and $N_1 = N_2$. In this case, $A = n_1/n_2 = \phi_1/\phi_2$. Unlike the parameter y, the phase behavior cannot be uniquely determined by the ratio of ϕ_1 and ϕ_2 in the homogeneous solution and we need to specify both ϕ_1 and ϕ_2. A three dimensional phase diagram is needed with, say, ϕ_1 in x-axis, ϕ_2 in y-axis, and ϕ_i in the z-axis. Some representative curves in 2D at given values of ϕ_2 are shown in Fig. 5 for counterions and salt ions being indistinguishable. It shows binodals for $\phi_1 = 0.02$ at different values of ϕ_2. For a complete description, similar phase diagrams at different values of ϕ_1 should be constructed. Instead of Fig. 5, a simple way to demonstrate the effect of asymmetry is a plot as shown in Fig. 6, where ϕ_{st} gives the amount of salt to be added to stabilize a homogeneous solution with the polycation and polyanion content of ϕ_1 and ϕ_2, respectively, against phase separation. In Fig. 6, data points of each color are threshold salt concentrations for a given ϕ_1. The plots show that with an

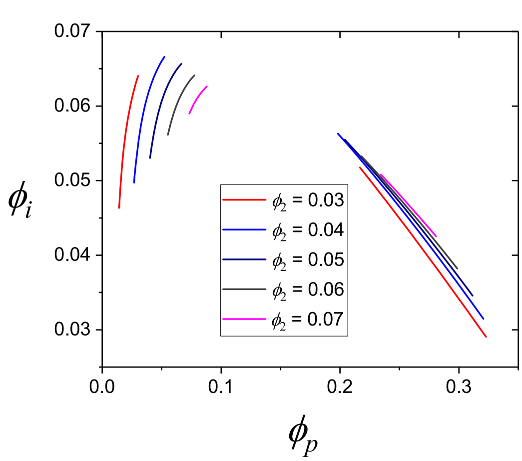

FIG. 5. Total ion (ϕ_i) vs polyelectrolyte (ϕ_p) concentration for an asymmetric solution of $N_1 = N_2 = 100$, $n_1 < n_2$ for $\epsilon = 80$, $a_\chi = 1.7$, and $p = l = 0.55$ nm at $t = 0.062\,39$ (25 °C). Curves represent binodals for different values of ϕ_2 from 0.03 to 0.07 where ϕ_1 is kept constant at 0.02. Counterions are treated to be indistinguishable to the salt ions.

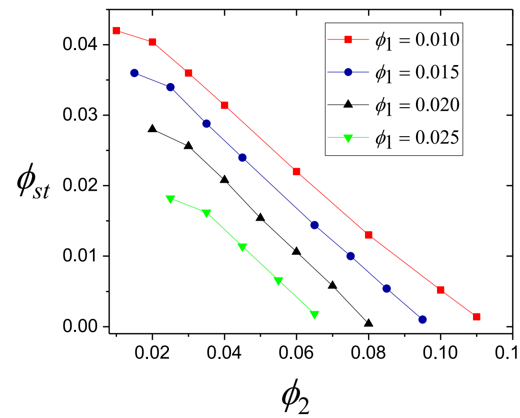

FIG. 6. Threshold salt concentration (ϕ_{st}) vs polyanion ϕ_2 concentration for an asymmetric solution of $N_1 = N_2 = 100$, $n_1 < n_2$ for $\epsilon = 80$, $a_\chi = 1.7$, and $p = l = 0.55$ nm at $t = 0.062\,39$ (25 °C). Each set of symbols represents a variation of ϕ_2 at fixed ϕ_1, where red squares, blue circles, black triangles, and green inverted triangles represent $\phi_1 = 0.010$, 0.015, 0.020, and 0.025, respectively. Counterions are treated to be indistinguishable to the salt ions.

increase in ϕ_2 (which increases polycation-polyanion composition asymmetry), the threshold amount of salt to be added decreases. This shows the suppressing effect of polycation-polyanion composition asymmetry on complex coacervation. The results in Figs. 4 and 6 show that complex coacervation is in general suppressed by any type of asymmetry in polycation and polyanion composition (either $y < 1$ or $n_1 < n_2$).

B. Solutions without counterions

Now we consider solutions without counterions which are relevant to the experiments in which counterions are washed out of the system before adding salt.[47] If the starting homogeneous solution is not symmetric in polycation-polyanion composition, we assume that there exist a minimum number of counterions required to make the solution electrically neutral.

1. *Effect of temperature and salt*

Figure 7 illustrates the binodals for salt versus polyelectrolyte concentrations at two different temperatures. The curves show a coexistence between a polyelectrolyte rich coacervate phase and a polyelectrolyte poor aqueous phase below a critical salt concentration (diamond). Like in the previous case of solutions with counterions, the phase diagrams show the suppression of complex coacervation by the addition of salt and the increase of temperature.

2. *Effect of chain length*

The effect of chain length on the coacervation phase behavior is demonstrated by the variation of critical points with chain length in Fig. 8. For a symmetric solution without counterions, it shows that with increasing polyelectrolyte chain length N, the critical salt concentration ϕ_{sc} increases, while the critical polyelectrolyte concentration ϕ_{pc} decreases. The increase in critical salt concentration with N demonstrates the promotion of complex coacervation by increasing polyelectrolyte chain length which is due to the entropic penalty of phase separation being lesser for longer chains.

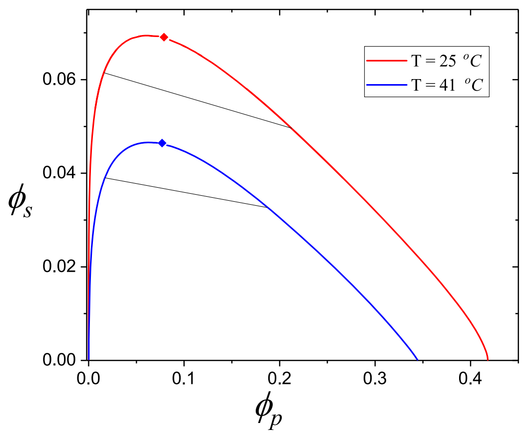

FIG. 7. Salt (ϕ_s) vs polyelectrolyte (ϕ_p) concentration for a symmetric solution with $N_1 = N_2 = 100$, $n_1 = n_2$ for $\epsilon = 80$, $a_\chi = 1.7$, and $p = l = 0.55$ nm. Curves represent binodals at $t = 0.062\,39$ (25 °C) (red, upper curve) and $t = 0.065\,70$ (41 °C) (blue, lower curve) including critical points (diamonds) and tie lines (black).

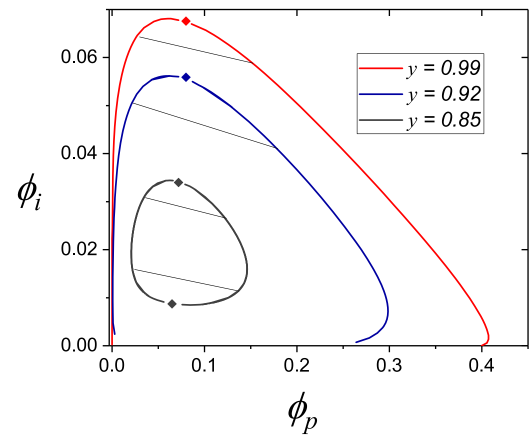

FIG. 9. Total ions (ϕ_i) vs polyelectrolyte (ϕ_p) for an asymmetric solution with $n_1 = n_2$, $N_1/N_2 = y$, $N_2 = 100$ for $\epsilon = 80$, $a_\chi = 1.7$, and $p = l = 0.55$ nm at $t = 0.062\,39$ (25 °C). Curves represent binodals for different degrees of asymmetry given by $y = 0.99$ (red, outer most curve), $y = 0.92$ (blue, middle curve), and $y = 0.85$ (gray, inner most curve), with tie lines (black) and critical points (diamonds). Since there are some counterions left to neutralize the asymmetric homogeneous solution, ϕ_i includes both salt and counterions. Salt and counterions are identical.

3. Effect of chain length asymmetry

We have investigated the influence of the chain length asymmetry in the phase behavior by considering a solution with $n_1 = n_2$ and varying the ratio of chain lengths of polycation and polyanion y at temperature $t = 0.062\,39$ (25 °C). The phase diagrams in Fig. 9 show how coacervation becomes less prevalent with decreasing symmetry, demonstrated by the shrinking region of instability. Upon decreasing the value of y, the area of the unstable region decreases. In fact, for the set of parameters used here, the area of the circular unstable region shrinks to zero and coacervation no longer occurs when the symmetry parameter y is below 0.835. The observed phase diagrams can be explained in terms of the un-complexed polyelectrolyte portion and the counterions which remain coupled with them to maintain electroneutrality. The shrinking of the unstable region with decreasing y is due to a reduction of dipole-dipole attractions and an increase of charge-charge repulsions. The existence of lower critical salt concentration originates from the entropy of ions; since some ions always remain coupled with the polyelectrolyte backbone to maintain electroneutrality, their entropy loss due to phase separation inhibits coacervation. The origin of the lower critical behavior is evident from a consideration of an asymmetric solution without salt. If its phase separates, creating a polymer-rich and a polymer-poor phase, the counterions should also partition to make each phase electrically neutral. The entropic penalty of partition of counterions stops the solution from phase separation. With the addition of salt, the partition of ions can be diluted and phase separation becomes possible.

Some experimental observations show that there is an additional precipitation below a threshold salt concentration[21,22,25] (presumably below our lower critical salt concentration). The present theory does not predict the existence of this precipitate phase at very low or no salt concentration. Nevertheless our predictions of a lower critical salt concentration and re-stabilization of the solution above an upper critical salt concentration are consistent with experiments.

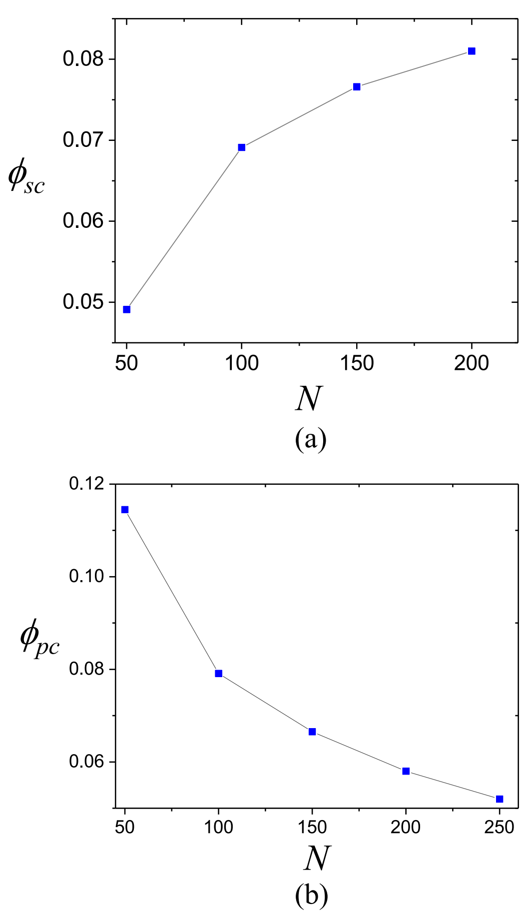

FIG. 8. (a) Critical salt concentration (ϕ_{sc}) vs chain length (N) and (b) critical polyelectrolyte concentration (ϕ_{pc}) vs chain length (N) for a symmetric solution of $N_1 = N_2 = N$, $n_1 = n_2$ for $\epsilon = 80$, $a_\chi = 1.7$, and $p = l = 0.55$ nm at $t = 0.062\,39$ (25 °C). Blue squares locate the calculated critical points and gray lines are for visual guide.

IV. CONCLUSION

By introducing the idea of electrostatic dipolar interactions between pair complexes, we have explored the effects of several experimentally relevant variables on complex coacervation. Common experimental observations including the promotion of complex coacervation by increasing chain length or by reducing temperature and the inhibition of coacervation by salt and by asymmetry in the concentrations of polycations and polyanions are predicted. In addition, we predict negative slopes for the tie lines in the coacervate phase diagrams (ϕ_i versus ϕ_p) due to the preferential partitioning of salt out of the dipole rich phase. The prediction of the negative slope of tie lines and the stabilization of the solution by polycation-polyanion composition asymmetry are also consistent with experiments. A quantitative comparison between the calculated phase diagram and experiments is not the objective of the present paper, as the particular values of the parameters of the model specific to the experimental systems are unknown. However, we expect that the phenomenon of enhancement of hydrophobicity by electrostatic interactions, negative slopes of tie lines, and the effect of asymmetry in complex coacervation phase behavior describe the general character of complex coacervation.

Our theory, being a mean field description, suffers from all the shortcomings of Flory-Huggins theory for polymer solutions and Debye-Hückel theory for strong electrolytes. In principle, the field theoretic model presented through Eqs. (11)–(20) can be solved using the numerical scheme of FTS[41–44] by including counterions and salt, to go beyond the present mean-field analytical results. Despite the possibility of various ways of complexation forming multiple trains of dipoles and multipoles, we have considered only the simplest case of pairwise ladderlike complexation. Nevertheless, this simple mean field treatment of complex coacervation captures the effect of electrostatic dipolar interactions arising due to the existence of soluble "small complexes" over a wide range of temperatures and salt concentrations to the phase behavior. Furthermore, the idea of considering dipole and multipole electrostatic interactions may have broad applicability to other charged polymeric systems such as polyzwitterions. Our theory can also be expanded and improved by considerations of other possible structures of complexation, multipole electrostatic interactions, the semi-flexible nature of charged polymer chains, the expanded range of polyelectrolyte-solvent hydrophobicity, and the correlations of dipolar interactions.

ACKNOWLEDGMENTS

We thank the National Science Foundation (No. DMR-1504265) and AFOSR Grant No. FA9550-17-1-0160 for financial support.

APPENDIX: CONSTRAINT EQUATIONS

The incompressibility condition for phase A (which coexists with phase B) is

$$[n_{1A}(N_1+N_2)+n_{2'A}N_2+n_{c1A}+n_{c2A}+n_{+A}+n_{-A}+n_{0A}]\frac{\ell^3}{V}=1, \tag{A1}$$

where the subscript A denotes that the various quantities are for phase A. In terms of volume fractions,

$$[\phi_{1A}(1+\frac{N_2}{N_1})+\phi_{2'A}+\phi_{c1A}+\phi_{c2A}+\phi_{+A}+\phi_{-A}+\phi_{0A}]=1. \tag{A2}$$

Analogously, for phase B, the incompressibility constraint is

$$[\phi_{1B}(1+\frac{N_2}{N_1})+\phi_{2'B}+\phi_{c1B}+\phi_{c2B}+\phi_{+B}+\phi_{-B}+\phi_{0B}]=1. \tag{A3}$$

The electroneutrality condition for phase A is

$$\alpha\phi_{1A}(\frac{N_2}{N_1}-1)+\alpha\phi_{2'A}+\phi_{c1A}+\phi_{-A}=\phi_{c2A}+\phi_{+A}. \tag{A4}$$

The lever rules are

$$\begin{aligned}
x\phi_{1A}+(1-x)\phi_{1B}&=\phi_1,\\
x\phi_{2'A}+(1-x)\phi_{2'B}&=\phi_{2'}=\phi_2-\phi_1\frac{N_2}{N_1},\\
x\phi_{c1A}+(1-x)\phi_{c1B}&=\phi_{c1}=\phi_1,\\
x\phi_{c2A}+(1-x)\phi_{c2B}&=\phi_{c2}=(1-\alpha)\phi_1+\alpha\phi_2,\\
x\phi_{+A}+(1-x)\phi_{+B}&=\frac{\phi_s}{2},\\
x\phi_{-A}+(1-x)\phi_{-B}&=\frac{\phi_s}{2},
\end{aligned} \tag{A5}$$

and the redundant lever rule for the solvent is

$$x\phi_{0A}+(1-x)\phi_{0B}=\phi_0.$$

Equations (A2)–(A5) are the nine independent constraints in the model. Therefore, the 15 variables are determined from these nine constraints by a six-dimensional minimization of the free energy for the phase separated system.

If the counterions and salt ions are indistinguishable, four-dimensional minimization is required as there are now five species (complexed polymer, un-complexed polyanion, cation, anion, and solvent) with 11 variables for two coexisting phases, and the seven constraints are given by

$$\begin{aligned}
&[\phi_{1A}(1+\frac{N_2}{N_1})+\phi_{2'A}+\phi_{+A}+\phi_{-A}+\phi_{0A}]=1,\\
&[\phi_{1B}(1+\frac{N_2}{N_1})+\phi_{2'B}+\phi_{+B}+\phi_{-B}+\phi_{0B}]=1,\\
&\alpha\phi_{1A}(\frac{N_2}{N_1}-1)+\alpha\phi_{2'A}+\phi_{-A}=\phi_{+A},\\
&x\phi_{1A}+(1-x)\phi_{1B}=\phi_1,\\
&x\phi_{2'A}+(1-x)\phi_{2'B}=\phi_{2'}=\phi_2-\phi_1\frac{N_2}{N_1},\\
&x\phi_{+A}+(1-x)\phi_{+B}=\frac{\phi_s}{2}+(1-\alpha)\phi_1+\alpha\phi_2,\\
&x\phi_{-A}+(1-x)\phi_{-B}=\frac{\phi_s}{2}+\phi_1.
\end{aligned} \tag{A6}$$

If the number concentrations of polycations and polyanions in the system are the same, then there is only a polymer species, namely, the complexed polyanion in the present model corresponding to either a diblock or triblock copolymer architecture. In this limit, when $N_1 \le N_2$ and the counterions and salt ions are distinguishable, there are 13 variables to be determined using the following 8 constraints:

$$
\begin{aligned}
&[\phi_{1A}(1+\frac{N_2}{N_1}) + \phi_{c1A} + \phi_{c2A} + \phi_{+A} + \phi_{-A} + \phi_{0A}] = 1,\\
&[\phi_{1B}(1+\frac{N_2}{N_1}) + \phi_{c1B} + \phi_{c2B} + \phi_{+B} + \phi_{-B} + \phi_{0B}] = 1,\\
&\alpha\phi_{1A}(\frac{N_2}{N_1} - 1) + \phi_{c1A} + \phi_{-A} = \phi_{c2A} + \phi_{+A},\\
&x\phi_{1A} + (1-x)\phi_{1B} = \phi_1,\\
&x\phi_{c1A} + (1-x)\phi_{c1B} = \phi_{c1} = \phi_1,\\
&x\phi_{c2A} + (1-x)\phi_{c2B} = \phi_{c2} = [1+\alpha(\frac{N_2}{N_1} - 1)]\phi_1,\\
&x\phi_{+A} + (1-x)\phi_{+B} = \frac{\phi_s}{2},\\
&x\phi_{-A} + (1-x)\phi_{-B} = \frac{\phi_s}{2}.
\end{aligned} \tag{A7}
$$

Therefore, a five-dimensional minimization of free energy needs to be used in the computation.

The computational effort is reduced to a three-dimensional minimization of free energy for the case of $n_1 = n_2$, namely, the two components of polycations and polyanions are imagined to constitute only one species and when the counterions and salt ions are indistinguishable. Now there are only four species (one polymer, cation, anion, and solvent) with nine independent variables to be determined by using the following six constraints:

$$
\begin{aligned}
&[\phi_{1A}(1+\frac{N_2}{N_1}) + \phi_{+A} + \phi_{-A} + \phi_{0A}] = 1,\\
&[\phi_{1B}(1+\frac{N_2}{N_1}) + \phi_{+B} + \phi_{-B} + \phi_{0B}] = 1,\\
&\alpha\phi_{1A}(\frac{N_2}{N_1} - 1) + \phi_{-A} = \phi_{+A},\\
&x\phi_{1A} + (1-x)\phi_{1B} = \phi_1,\\
&x\phi_{+A} + (1-x)\phi_{+B} = \frac{\phi_s}{2} + [1+\alpha(\frac{N_2}{N_1} - 1)]\phi_1,\\
&x\phi_{-A} + (1-x)\phi_{-B} = \frac{\phi_s}{2} + \phi_1.
\end{aligned} \tag{A8}
$$

Thus the number of dimensions required for free energy minimization depends on whether $n_1 = n_2$ or not and on whether the salt ions and counterions are distinguishable or not. For $n_1 \neq n_2$, the number of dimensions for minimization is 6 for distinguishable ions and 4 for indistinguishable ions. For $n_1 = n_2$, the number of dimensions for minimization is 5 for distinguishable ions and 3 for indistinguishable ions. We provide typical examples for both $n_1 = n_2$ and $n_1 \neq n_2$ and for both distinguishable and indistinguishable counterions and salt ions.

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